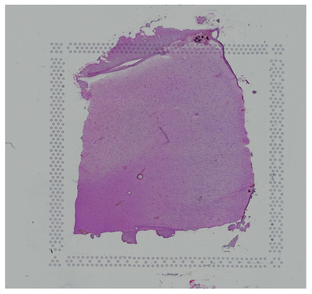
A

Spatial transcriptomics



Features

Spots

Feature matrix

PCA

Features

Spots

Feature matrix

Spatial coordinates

KNN

Adjacency matrix

B

View1

Feature matrix X1

Adjacency A1

View2

Feature matrix X2

Adjacency A2

C

Feature matrix X1

Adjacency A1

Encoder

Shared weight

Feature matrix X2

Adjacency A2

Encoder

Dyt Module

Z1

Dyt Module

Z2

Loss ali

Similarity Matrix S

Sample Weight Calculator

WeightVector W

Weighted Total Loss

Aij= 1 w Neighbor

Attract neighbors (enhance similarity)

Pull neighbors together

Aij= 0 w Non Neighbor

Filter non-neighbors (suppress false similarity)

Suppress false similarity

Spatial filter A

Confidence weighting(sample weight w)

Pairwise loss

D

